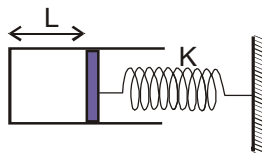


**PHYSICS**

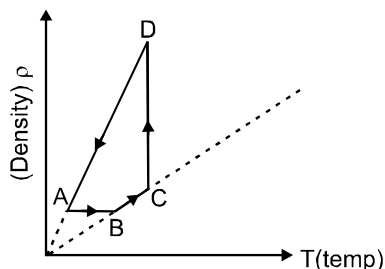
TARGET : JEE- Advanced

**CAPS-20****KTG & THERMODYNAMIC****SCQ (Single Correct Type) :**

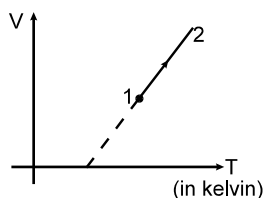
1. A fixed container is fitted with a piston which is attached to a spring of spring constant  $k$ . The other end of the spring is attached to a rigid wall. Initially the spring is in its natural length and the length of container between the piston and its side wall is  $L$ . Now an ideal diatomic gas is slowly filled in the container so that the piston moves quasistatically. It pushed the piston by  $x$  so that the spring now is compressed by  $x$ . The total rotational kinetic energy of the gas molecules in terms of the displacement  $x$  of the piston is (there is vacuum outside the container)



- (A)  $k \times L$                       (B)  $4k \times L$                       (C)  $kx(x+L)$                       (D)  $\frac{2kx^2}{L}$
2. Consider the following cyclic process for mono-atomic ideal gas. This process consist of four process AB, BC, CD and DA. The number of processes out of these four processes in which heat is supplied to the gas is :

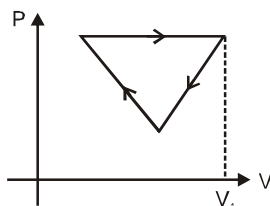


- (A) 1                      (B) 2                      (C) 3                      (D) 4
3. V-T diagram for a process of a given mass of ideal gas is as shown in the figure. During the process pressure of gas.



- (A) first increases then decreases                      (B) continuously decreases  
(C) continuously increases                      (D) first decreases then increases.

4. An ideal gas at temperature 'T' follows the law  $P^2 = V$  where P is pressure and V is volume of gas, Then the volume expansion co-efficient of gas is.
- (A)  $\frac{1}{T}$  (B)  $\frac{2}{T}$  (C)  $\frac{3}{T}$  (D)  $\frac{2}{3T}$
5. The diagram shown an equilateral cyclic process on P-V diagram. The ratio of maximum pressure to minimum pressure is 2. The maximum temperature in the process is  $T_1$ . The work done by the two moles of ideal gas in the cyclic process is equal to (R is universal gas constant)-



- (A)  $\frac{R^2 T_1^2}{4\sqrt{3}V_1^2}$  (B)  $\frac{4R^2 T_1^2}{\sqrt{3}V_1^2}$  (C)  $\frac{R^2 T_1^2}{\sqrt{3}V_1^2}$  (D)  $\frac{2}{\sqrt{3}} \frac{R^2 T_1^2}{V_1^2}$
6. 56 gm of  $N_2$  gas is passed through a reversible process in which internal energy changes with pressure (P) according to  $U \propto P^2$ . Initial temperature of the gas is  $T_0$ . During the process, if volume of the gas is doubled, then heat given to the gas will be :
- (A)  $3RT_0$  (B)  $16T_0$  (C)  $9RT_0$  (D)  $18RT_0$

**MCQ (One or more than one correct) :**

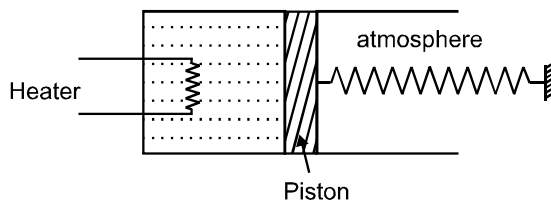
7. A certain thermodynamic cycle is given by the equation

$$2\left(\frac{P-P_0}{P_0}\right)^2 + 2\left(\frac{V-V_0}{V_0}\right)^2 = 1$$

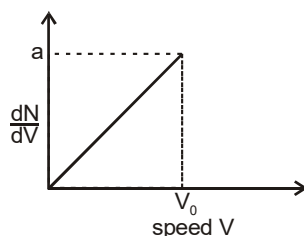
where P and V represent the pressure and the volume of the ideal gas involved. If  $P_0$  and  $V_0$  are both expressed in SI units then their values are equal. The cycle is clockwise when represented on the P-V diagram and the number of moles of the gas is n. Now choose the correct statements:

- (A) The maximum temperature obtained in the cycle is  $\frac{9V_0 P_0}{4nR}$
- (B) The minimum temperature obtained in the cycle is  $\frac{V_0 P_0}{4nR}$
- (C) The total work done by the gas in one cycle is  $\frac{\pi V_0 P_0}{2}$
- (D) The maximum pressure obtained in the cycle is  $P_0 \left[1 + \frac{1}{\sqrt{2}}\right]$

8. An ideal gas is filled in a cylinder as shown in figure. The initial temperature, pressure and volume of the gas are  $T_0$ ,  $P_0$  and  $V_0$  respectively where  $P_0 =$  atmospheric pressure. A light and smooth piston of area  $A$  is connected to a spring of spring constant  $K$ , which is initially in natural length. Now the gas is heated slowly for some time, due to which the piston moves out slowly by a distance 'x'. Then :



- (A) Final pressure of the gas is  $P_0 + \frac{Kx}{A}$
- (B) Final temperature of the gas is  $\left(1 + \frac{Kx}{P_0 A}\right) \left(1 + \frac{Ax}{V_0}\right) T_0$
- (C) The gas is undergoing constant pressure process
- (D) Work done by the gas is  $\frac{1}{2} Kx^2$
9. Graph shows a hypothetical speed distribution for a sample of  $N$  gas particle (for  $V > V_0$ ;  $\frac{dN}{dV} = 0$ ),  $\frac{dN}{dV}$  is rate of change of number of particles with change velocity

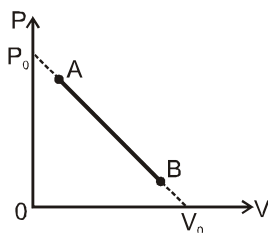


- (A) The value of  $aV_0$  is  $2N$ .
- (B) The ratio  $V_{avg}/V_0$  is equal to  $2/3$ .
- (C) The ratio  $V_{rms}/V_0$  is equal to  $1/\sqrt{2}$ .
- (D) Three fourth of the total particle has a speed between  $0.5 V_0$  and  $V_0$ .
10. One mole of an ideal gas undergoes a process such that  $P \propto \frac{1}{\sqrt{T}}$ . The molar heat capacity of this process is  $4R$ ,  $R =$  universal gas constant.
- (A) The work done by the gas is  $1.5 R\Delta T$  where  $\Delta T$  is the change in temperature
- (B) Degree of freedom of the gas is 5.
- (C) On increase of temperature, volume increases
- (D) On increase of temperature, volume decreases

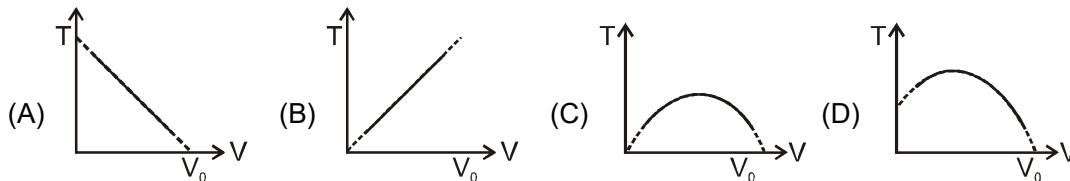
11. An ideal diatomic gas undergoes a process in which its internal energy ( $U$ ) relates to the volume ( $V$ ) as  $U = \alpha\sqrt{V}$ , here  $\alpha$  is a constant. The internal energy of gas is increased by 100 J.
- (A) The work performed by gas is 80 J.
- (B) The amount of heat to be transferred to gas is 180 J.
- (C) The molar specific heat of gas is  $\frac{9R}{2}$
- (D) The molar specific heat of gas is  $\frac{5R}{2}$
12. One mole of an ideal monatomic gas is taken from temperature  $T_0$  to  $2T_0$  by the process  $T^4P^{-1} = \text{constant}$ .
- (A) The molar specific heat of gas is  $-\frac{3R}{2}$  (B) The molar specific heat of gas is  $\frac{3R}{2}$
- (C) Work done by gas is  $-3RT_0$  (D) Work done by gas is  $3RT_0$
13. An ideal gas is taken from the state A (pressure  $P$ , volume  $V$ ) to the state B (pressure  $P/2$ , volume  $2V$ ) along a straight line path in the  $P$ - $V$  diagram. Select the correct statement(s) from the following :
- (A) The work done by the gas in the process A to B exceeds the work that would be done by it if the system were taken from A to B along the isotherm.
- (B) In the  $T$ - $V$  diagram, the path AB becomes a part of the parabola.
- (C) In the  $P$ - $T$  diagram, the path AB becomes a part of hyperbola.
- (D) In going from A to B, the temperature  $T$  of the gas first increases to a maximum value and then decreases.

### Comprehension Type Question:

One mole of an ideal monatomic gas undergoes a linear process from A to B, in which is pressure  $P$  and its volume  $V$  change as shown in figure



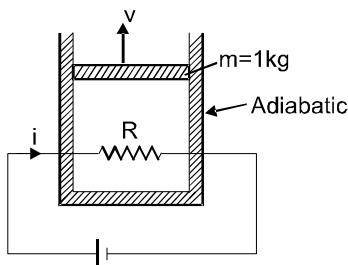
14. The absolute temperature  $T$  versus volume  $V$  for the given process is



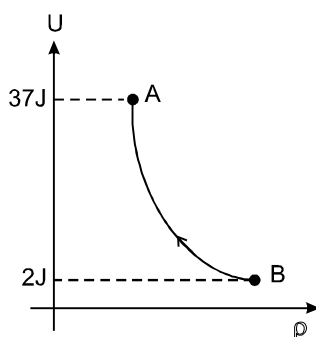
15. The maximum temperature of the gas during this process is  
 (A)  $\frac{P_0 V_0}{2 R}$  (B)  $\frac{P_0 V_0}{4 R}$  (C)  $\frac{3 P_0 V_0}{4 R}$  (D)  $\frac{3 P_0 V_0}{2 R}$
16. As the volume of the gas is increased, in some range of volume the gas expands with absorbing the heat (the endothermic process) ; in the other range the gas emits the heat (the exothermic process). Then the volume after which if the volume of gas is further increased the given process switches from endothermic to exothermic is  
 (A)  $\frac{2V_0}{8}$  (B)  $\frac{3V_0}{8}$  (C)  $\frac{5V_0}{8}$  (D) none of these

### Numerical based Questions :

17. Two identical containers A and B with frictionless pistons contain the same ideal gas at the same temperature and the same volume  $V$ . The mass of the gas in A is  $m_A$  and that in the B is  $m_B$ . The gas in each cylinder is now allowed to expand isothermally to the same final volume  $2V$ . The changes in the pressure in A and B are found to be  $\Delta P$  and  $1.5\Delta P$  respectively. Then find  $\frac{14m_B}{m_A}$ .
18. Current  $i = 2A$  flows through the resistance  $R = 10\Omega$ . With what constant speed  $v$  (in m/s), must the piston move in upward direction so that temperature of ideal gas may remain unchanged. ( $g = 10 \text{ m/s}^2$ )



19. Figure shows the variation of internal energy " $U$ " with the density " $\rho$ " of one mole of ideal diatomic gas. Process BA is a part of rectangular hyperbola. If the work done by gas in the process BA is  $W$  joules. Find  $W$  ?



**Matrix Match Type :**

20. An ideal monoatomic gas undergoes different types of processes which are described in column-I. Match the corresponding effects in column-II. The letters have usual meaning.

**Column-I**

(A)  $P = 2V^2$

(B)  $PV^2 = \text{constant}$

(C)  $C = C_V + 2R$

(D)  $C = C_V - 2R$

**Column-II**

(p) If volume increases then temperature will also increase.

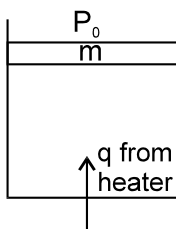
(q) If volume increases then temperature will decrease.

(r) For expansion, heat will have to be supplied to the gas.

(s) If temperature increases then work done by gas is positive.

**Subjective Type Questions :**

21. Two moles of an ideal monoatomic gas are contained in a vertical cylinder of cross sectional area  $A$  as shown in the figure. The piston is frictionless and has a mass  $m$ . At a certain instant a heater starts supplying heat to the gas at a constant rate  $q$  J/s. Find the steady velocity of the piston under isobaric condition. All the boundaries are thermally insulated.



22. A piston can freely move inside a horizontal cylinder closed from both ends. Initially, the piston separates the inside space of the cylinder into two equal parts each of volume  $V_0$ , in which an ideal gas is contained under the same pressure  $p_0$  and at the same temperature. What work has to be performed in order to increase isothermally the volume of one part of gas  $\eta$  times compared to that of the other by slowly moving the piston?
23. Two moles of an ideal monoatomic gas are confined within a cylinder by a massless and frictionless spring loaded piston of cross-sectional area  $4 \times 10^{-3} \text{ m}^2$ . The spring is, initially in its relaxed state. Now the gas is heated by an electric heater, placed inside the cylinder, for some time. During this time, the gas expands and does 50 J of work in moving the piston through a distance 0.10 m. The temperature of the gas increases by 50 K. Calculate the spring constant and the heat supplied by the heater.  $P_{\text{atm}} = 1 \times 10^5 \text{ N/m}^2$ ,  $R = 8.314 \text{ J/mol-K}$

24. Two vessels A and B, thermally insulated, contain an ideal monoatomic gas. A small tube fitted with a valve connects these vessels. Initially the vessel A has 2 liters of gas at 300 K and  $2 \times 10^5 \text{ N m}^{-2}$  pressure while vessel B has 4 liters of gas at 350 K and  $4 \times 10^5 \text{ N m}^{-2}$  pressure. The valve is now opened and the system reaches equilibrium in pressure and temperature. Calculate the new pressure and temperature. ( $R = \frac{25}{3} \text{ J/mol-K}$ )
25. An ideal gas ( $C_p/C_v = \gamma$ ) having initial pressure  $P_0$  and volume  $V_0$ . (a) The gas is taken isothermally to a pressure  $2P_0$  and then adiabatically to a pressure  $4P_0$ . Find the final volume. (b) The gas is brought back to its initial state. It is adiabatically taken to a pressure  $2P_0$  and then isothermally to a pressure  $4P_0$ . Find the final volume.

**ANSWER KEY (CAPS-20)**

- |                                                                                                              |                                                             |                                              |           |           |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------|-----------|-----------|
| 1. (C)                                                                                                       | 2. (B)                                                      | 3. (B)                                       | 4. (D)    | 5. (C)    |
| 6. (D)                                                                                                       | 7. (ABCD)                                                   | 8. (AB)                                      | 9. (ABCD) | 10. (ABC) |
| 11. (ABC)                                                                                                    | 12. (AC)                                                    | 13. (ABD)                                    | 14. (C)   | 15. (B)   |
| 16. (C)                                                                                                      | 17. 21                                                      | 18. $v = 4 \text{ m/s}$                      | 19. 14    |           |
| 20. (A) p,r,s (B) q (C) p,r,s (D) q,r                                                                        | 21. $\frac{2q}{5(mg + P_0 A)}$                              | 22. $W = p_0 V_0 \ln [(\eta + 1)^2 / 4\eta]$ |           |           |
| 23. $K = 2000 \text{ N/m}$ , $Q = 1297 \text{ Joules approx.}$                                               |                                                             |                                              |           |           |
| 24. $P = \frac{10}{3} \times 10^5 \text{ N/m}^2$ , $T = \frac{10500}{31} \text{ K} \approx 338.71 \text{ K}$ | 25. $\frac{V_0}{2^{\frac{\gamma+1}{\gamma}}}$ in each cases |                                              |           |           |